# 20.2 Other Inclusion Regions

The Gershgorin discs of §01 come from one row of $\mathbf{Ax} = \lambda\mathbf{x}$, read at the largest coordinate of $\mathbf{x}$. That leaves two kinds of information unused. The first is the second largest coordinate, which the triangle inequality simply replaced by the largest; using it gives a region of ovals that is never worse than the discs. The second is the geometry of the inner product, which the discs ignore entirely: the number $\mathbf{x}^*\mathbf{Ax}$ over unit vectors $\mathbf{x}$ fills a region of the plane, the numerical range, that contains every eigenvalue. This section builds both regions. Its main theorem is that the numerical range is always convex. It ends with a rectangle for the spectrum whose sides are eigenvalues of Hermitian matrices, and with a warning about what these regions cannot do.

**Throughout, matrices are complex and** $n \geq 1$**,** and $\|\cdot\|$ on $\mathbb{C}^n$ is the Euclidean norm. As in §01, $r_i(\mathbf{A}) = \sum_{j\neq i}|a_{ij}|$ is the $i$-th Gershgorin radius and $D_i(\mathbf{A})$ the $i$-th disc (Definition 20.1.1). The plane $\mathbb{C}$ is treated as a real vector space when convexity is discussed, as Chapter 18 §01 does.

## 20.2.1 Using two coordinates: Brauer's ovals

In the proof of Theorem 20.1.1 every $|x_j|$ with $j \neq k$ was bounded by the largest one, $|x_k|$. That is wasteful when the second largest coordinate is much smaller. So keep two indices: $p$, where $|x_p|$ is largest, and $q \neq p$, where it is largest among the rest. Row $p$ bounds $|\lambda - a_{pp}||x_p|$ by $r_p|x_q|$, since every other coordinate is at most $|x_q|$. Row $q$ bounds $|\lambda - a_{qq}||x_q|$ by $r_q|x_p|$. Multiplying the two bounds cancels the unknown coordinates. The result is a region defined by a product of distances.

**Theorem 20.2.1. Brauer's Ovals of Cassini**

Let $n \geq 2$ and $\mathbf{A} \in M_n(\mathbb{C})$. For $i \neq j$ put

$$O_{ij}(\mathbf{A}) \coloneqq \{\, z \in \mathbb{C} : |z - a_{ii}|\,|z - a_{jj}| \leq r_i(\mathbf{A})\, r_j(\mathbf{A}) \,\}.$$

(a) Every eigenvalue of $\mathbf{A}$ lies in $\bigcup_{i\neq j} O_{ij}(\mathbf{A})$.

(b) $O_{ij}(\mathbf{A}) \subseteq D_i(\mathbf{A}) \cup D_j(\mathbf{A})$ for every $i \neq j$. Hence $\bigcup_{i\neq j} O_{ij}(\mathbf{A}) \subseteq \bigcup_i D_i(\mathbf{A})$.

*Idea.* For (a), take the two largest coordinates of an eigenvector as above and multiply the two row estimates. One case needs separate care: if the second largest coordinate is 0, there is nothing to divide by, but then $\mathbf{x}$ is a multiple of $\mathbf{e}_p$ and $\lambda = a_{pp}$, which lies in every oval through $a_{pp}$. For (b), a point outside both discs is farther than $r_i$ from $a_{ii}$ and farther than $r_j$ from $a_{jj}$, so the product of its distances exceeds $r_i r_j$.

*Proof.*

(a) Let $\mathbf{Ax} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$. Choose $p$ with $|x_p| = \max_j |x_j|$, which is positive, and then $q \neq p$ with $|x_q| = \max_{j\neq p}|x_j|$; this is possible because $n \geq 2$.
*Case 1:* $x_q = 0$. Then $x_j = 0$ for every $j \neq p$, and row $p$ of $\mathbf{Ax} = \lambda\mathbf{x}$ reads $a_{pp}x_p = \lambda x_p$. As $x_p \neq 0$, $\lambda = a_{pp}$. The left side of the inequality defining $O_{pq}(\mathbf{A})$ is then 0, and the right side is $\geq 0$, so $\lambda \in O_{pq}(\mathbf{A})$.
*Case 2:* $x_q \neq 0$. Row $p$ gives $(\lambda - a_{pp})x_p = \sum_{j\neq p} a_{pj}x_j$. Every $j \neq p$ has $|x_j| \leq |x_q|$, so

by the triangle inequality (Theorem 0.6.4)

$$|\lambda - a_{pp}|\,|x_p| \le r_p(\mathbf{A})\,|x_q|.$$

Row $q$ gives $(\lambda - a_{qq})x_q = \sum_{j\neq q} a_{qj}x_j$. Every $j$ has $|x_j| \le |x_p|$, so

$$|\lambda - a_{qq}|\,|x_q| \le r_q(\mathbf{A})\,|x_p|.$$

Both sides of both inequalities are nonnegative, so they may be multiplied:

$$|\lambda - a_{pp}|\,|\lambda - a_{qq}|\,|x_p|\,|x_q| \le r_p(\mathbf{A})\,r_q(\mathbf{A})\,|x_p|\,|x_q|.$$

Dividing by $|x_p||x_q| > 0$ gives $\lambda \in O_{pq}(\mathbf{A})$.

(b) Let $z \notin D_i(\mathbf{A}) \cup D_j(\mathbf{A})$, so $a := |z - a_{ii}| > r_i(\mathbf{A}) \ge 0$ and $c := |z - a_{jj}| > r_j(\mathbf{A}) \ge 0$. Since $c > 0$ and $r_i(\mathbf{A}) \ge 0$,

$$ac > r_i(\mathbf{A})\,c \ge r_i(\mathbf{A})\,r_j(\mathbf{A}),$$

so $z \notin O_{ij}(\mathbf{A})$. Taking complements, $O_{ij}(\mathbf{A}) \subseteq D_i(\mathbf{A}) \cup D_j(\mathbf{A})$, and taking the union over all pairs gives the last claim. This proves the theorem.

■

The sets $O_{ij}(\mathbf{A})$ are the **ovals of Cassini**: the points whose distances to two fixed foci have product at most a constant. Since $O_{ij}(\mathbf{A}) = O_{ji}(\mathbf{A})$, there are $n(n-1)/2$ of them to draw, against $n$ discs. The hypothesis $n \ge 2$ is there because an oval needs two distinct indices; for $n = 1$ there are none, and the eigenvalue $a_{11}$ must be found some other way. For $n = 2$ the single oval is as good as a region can be. The eigenvalues satisfy $(\lambda - a_{11})(\lambda - a_{22}) = a_{12}a_{21}$, so taking moduli, $|\lambda - a_{11}||\lambda - a_{22}| = r_1 r_2$: both eigenvalues lie on the boundary curve of $O_{12}$.

**Example. Ovals where discs fail**

Let $\mathbf{A} = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 1 & 1 & 1 \end{pmatrix}$. Show that the Gershgorin discs cannot prove $\mathbf{A}$ invertible, and that the ovals can.

*Solution.* The radii are $r_1 = 1$, $r_2 = 1$, $r_3 = 2$, and $D_3(\mathbf{A}) = \{|z - 1| \le 2\}$ contains 0. So Theorem 20.1.1 (a) allows the eigenvalue 0. The column radii are $1, 2, 1$, and the second column disc $|z - 4| \le 2$ misses 0, but the third, $|z - 1| \le 1$, contains it. So the discs cannot decide.

Now test $z = 0$ against the three ovals:

$$\begin{aligned} O_{12}:&\ |0-4|\,|0-4| = 16 > 1 = r_1 r_2, \\ O_{13}:&\ |0-4|\,|0-1| = 4 > 2 = r_1 r_3, \\ O_{23}:&\ |0-4|\,|0-1| = 4 > 2 = r_2 r_3. \end{aligned}$$

So 0 lies in no oval, it is not an eigenvalue by Theorem 20.2.1 (a), and $\mathbf{A}$ is invertible by Theorem 9.2.1. Indeed, expanding along the first row, $\det \mathbf{A} = 4(4-1) - 1(0-1) = 13$. The ovals also exclude the point $-1$, which lies on the boundary of $D_3(\mathbf{A})$: there $|-1-4|\,|-1-4| =$

$25 > 1$ rules out $O_{12}$, and $|-1-4|\,|-1-1| = 10 > 2$ rules out $O_{13}$ and $O_{23}$.

The same test gives a criterion in the spirit of Corollary 20.1.2: if $n \geq 2$, as in Theorem 20.2.1, and $|a_{ii}|\,|a_{jj}| > r_i(\mathbf{A})\,r_j(\mathbf{A})$ for **every** pair $i \neq j$, then 0 lies in no oval and $\mathbf{A}$ is invertible. One row may fail to dominate, as row 3 did above, provided every other row dominates enough to compensate.

*Quick check.* Why does the proof of Theorem 20.2.1 (a) choose $q$ to maximize $|x_q|$ among the indices other than $p$, rather than taking any $q \neq p$?

*Solution.* The estimate from row $p$ bounds every $|x_j|$, $j \neq p$, by $|x_q|$. That is true only if $|x_q|$ is the largest of them. With an arbitrary $q$ the row-$p$ estimate could fail, and the product of the two estimates would no longer cancel down to $r_p r_q$.

## 20.2.2 The numerical range

Now a different source of information. For a Hermitian $\mathbf{A}$, Chapter 17 §01 studied the Rayleigh quotient, and Proposition 17.1.1 showed that its values $\mathbf{x}^*\mathbf{A}\mathbf{x}$ on unit vectors fill exactly the interval $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$, which contains every eigenvalue. The number $\mathbf{x}^*\mathbf{A}\mathbf{x} = \langle \mathbf{A}\mathbf{x}, \mathbf{x}\rangle$ makes sense for every square matrix. Without the Hermitian hypothesis it is complex, and its values fill a region of the plane rather than an interval.

*The numerical range is the set of all values $\mathbf{x}^*\mathbf{A}\mathbf{x}$ as $\mathbf{x}$ runs over the unit vectors.*

**Definition 20.2.1. Numerical Range**

Let $\mathbf{A} \in M_n(\mathbb{C})$. The **numerical range** of $\mathbf{A}$ is the subset of $\mathbb{C}$

$$W(\mathbf{A}) := \{\, \mathbf{x}^*\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{C}^n,\ \|\mathbf{x}\| = 1 \,\}.$$

In words: feed $\mathbf{A}$ a vector of length **exactly** 1, take the inner product of the output with the input, and collect all the complex numbers that arise. The vectors range over **complex** unit vectors, even when $\mathbf{A}$ is real, and the conjugate in $\mathbf{x}^*$ is part of the definition. $W(\mathbf{A})$ is always nonempty, since $\mathbf{e}_1$ is a unit vector, and $\mathbf{e}_i^*\mathbf{A}\mathbf{e}_i = a_{ii}$, so every diagonal entry lies in $W(\mathbf{A})$. The notation always carries its argument: $W(\mathbf{A})$, never $W$ alone, since $W$ also names subspaces.

Examples, simplest first.

- $n = 1$**, and scalar matrices.** For $\mathbf{A} = (a)$ the unit vectors are the scalars $x$ with $|x| = 1$, and $\overline{x}ax = a|x|^2 = a$. So $W((a)) = \{a\}$. Likewise $W(c\mathbf{I}) = \{c\}$, since $\mathbf{x}^*(c\mathbf{I})\mathbf{x} = c\|\mathbf{x}\|^2 = c$.
- **Diagonal matrices.** For $\mathbf{A} = \operatorname{diag}(d_1, \ldots, d_n)$, $\mathbf{x}^*\mathbf{A}\mathbf{x} = \sum_i d_i|x_i|^2$. The weights $t_i = |x_i|^2$ are nonnegative with sum $\|\mathbf{x}\|^2 = 1$, and every such list of weights arises, from $x_i = \sqrt{t_i}$. So $W(\mathbf{A})$ is the set of all weighted averages of $d_1, \ldots, d_n$, which is $\operatorname{conv}\{d_1, \ldots, d_n\}$ by Theorem 18.1.3. For $\operatorname{diag}(1, i, -1)$ it is the filled triangle with these three vertices.
- **Hermitian matrices.** By Proposition 17.1.1, $W(\mathbf{A}) = [\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$, a segment of the real axis.
- **A Jordan block.** The next example computes $W(\mathbf{J}_2(0))$, and it is a whole disc, although the only eigenvalue is 0.

**Example. The numerical range of a Jordan block**

Show that $W(\mathbf{J}_2(0)) = \{z \in \mathbb{C} : |z| \le \frac{1}{2}\}$.

*Solution.* Write $\mathbf{x} = (x_1, x_2)$ with $|x_1|^2 + |x_2|^2 = 1$. Then $\mathbf{J}_2(0)\mathbf{x} = (x_2, 0)$, so $\mathbf{x}^*\mathbf{J}_2(0)\mathbf{x} = \overline{x_1}\, x_2$.

($\subseteq$) Since $0 \le (|x_1| - |x_2|)^2 = 1 - 2|x_1||x_2|$, we get $|\overline{x_1}x_2| = |x_1||x_2| \le \frac{1}{2}$.

($\supseteq$) Let $z = \rho\,\omega$ with $0 \le \rho \le \frac{1}{2}$ and $|\omega| = 1$. Since $1 - 4\rho^2 \ge 0$, we may put

$$a = \sqrt{\tfrac{1+\sqrt{1-4\rho^2}}{2}}, \qquad b = \sqrt{\tfrac{1-\sqrt{1-4\rho^2}}{2}},$$

both real and nonnegative. Then $a^2 + b^2 = 1$ and $a^2b^2 = \frac{1}{4}(1 - (1 - 4\rho^2)) = \rho^2$, so $ab = \rho$. The unit vector $\mathbf{x} = (a, b\,\omega)$ gives $\mathbf{x}^*\mathbf{J}_2(0)\mathbf{x} = a\,b\,\omega = z$. Hence $W(\mathbf{J}_2(0))$ is the closed disc of radius $\frac{1}{2}$ about 0.

**Nonexample by minimal change.** Drop the conjugate, and collect the numbers $\mathbf{x}^\top\mathbf{A}\mathbf{x}$ for unit $\mathbf{x}$ instead. For $\mathbf{A} = \mathbf{I}_2$ and the unit vector $\mathbf{x} = \frac{1}{\sqrt{2}}(1, i)$, $\mathbf{x}^\top\mathbf{I}_2\mathbf{x} = \frac{1}{2}(1 + i^2) = 0$, while $\mathbf{x}^*\mathbf{I}_2\mathbf{x} = 1$. The modified set for $\mathbf{I}_2$ contains 0, which is not an eigenvalue, while $W(\mathbf{I}_2) = \{1\}$ is exactly the spectrum. What breaks is that $\mathbf{x}^\top\mathbf{x}$ is not $\|\mathbf{x}\|^2$ over $\mathbb{C}$. The conjugate is what makes $\mathbf{x}^*(c\mathbf{I})\mathbf{x} = c$ on the unit sphere, and every property below leans on that.

**Why this definition.** The unit length plays the role that division by $\|\mathbf{x}\|^2$ played for the Rayleigh quotient. Without it the values $\mathbf{x}^*\mathbf{A}\mathbf{x}$ over all $\mathbf{x}$ form a set closed under multiplication by positive reals, which is all of $\mathbb{C}$ or a cone, and records nothing about the size of $\mathbf{A}$. With it, the set is bounded, and the eigenvalues are in it.

**Proposition 20.2.2. Basic Properties of the Numerical Range**

Let $\mathbf{A} \in M_n(\mathbb{C})$.

(a) $\operatorname{spec}(\mathbf{A}) \subseteq W(\mathbf{A})$.

(b) $W(\alpha\mathbf{A} + \beta\mathbf{I}) = \alpha W(\mathbf{A}) + \beta \coloneqq \{\alpha w + \beta : w \in W(\mathbf{A})\}$ for all $\alpha, \beta \in \mathbb{C}$.

(c) $W(\mathbf{U}^*\mathbf{A}\mathbf{U}) = W(\mathbf{A})$ for every unitary $\mathbf{U} \in M_n(\mathbb{C})$.

(d) $W(\mathbf{A})$ is compact: every sequence in $W(\mathbf{A})$ has a subsequence converging to a point of $W(\mathbf{A})$. In particular $W(\mathbf{A})$ is closed.

(e) $W(\mathbf{A}) \subseteq \{z : |z| \le \|\mathbf{A}\|_2\}$.

*Idea.* Four of the five parts are one line of algebra with a unit vector: evaluate at a unit eigenvector for (a), push the scalars or the unitary through $\mathbf{x}^*(\cdot)\mathbf{x}$ for (b) and (c), and apply Cauchy–Schwarz for (e). Only (d) needs analysis, and there the point is that $W(\mathbf{A})$ is the image of the unit sphere under a continuous map, so the compactness of the sphere passes to it.

*Proof.*

(a) Let $\lambda \in \operatorname{spec}(\mathbf{A})$, with eigenvector $\mathbf{v}$. Then $\mathbf{x} = \mathbf{v}/\|\mathbf{v}\|$ is a unit eigenvector, and $\mathbf{x}^*\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}^*\mathbf{x} = \lambda$.

(b) For a unit $\mathbf{x}$, $\mathbf{x}^*(\alpha\mathbf{A} + \beta\mathbf{I})\mathbf{x} = \alpha\,\mathbf{x}^*\mathbf{A}\mathbf{x} + \beta\,\mathbf{x}^*\mathbf{x} = \alpha\,\mathbf{x}^*\mathbf{A}\mathbf{x} + \beta$. As $\mathbf{x}$ runs over the unit vectors, the left side runs over $W(\alpha\mathbf{A} + \beta\mathbf{I})$ and the right side over $\alpha W(\mathbf{A}) + \beta$.

(c) For a unit $\mathbf{x}$, $\mathbf{x}^*\mathbf{U}^*\mathbf{A}\mathbf{U}\mathbf{x} = (\mathbf{U}\mathbf{x})^*\mathbf{A}(\mathbf{U}\mathbf{x})$, and $\mathbf{U}\mathbf{x}$ is a unit vector because $\mathbf{U}$ is an isometry (Theorem 11.7.3). So $W(\mathbf{U}^*\mathbf{A}\mathbf{U}) \subseteq W(\mathbf{A})$. Conversely, every unit $\mathbf{y}$ equals $\mathbf{U}\mathbf{x}$ for the unit vector $\mathbf{x} = \mathbf{U}^*\mathbf{y}$, so every value $\mathbf{y}^*\mathbf{A}\mathbf{y}$ arises, and the two sets are equal.

(d) First, $f(\mathbf{x}) = \mathbf{x}^*\mathbf{A}\mathbf{x}$ is continuous on the unit sphere. For unit vectors $\mathbf{x}, \mathbf{y}$,

$$\begin{aligned} |f(\mathbf{x}) - f(\mathbf{y})| &= |\mathbf{x}^*\mathbf{A}(\mathbf{x} - \mathbf{y}) + (\mathbf{x} - \mathbf{y})^*\mathbf{A}\mathbf{y}| \\ &\le 2\,\|\mathbf{A}\|_2\,\|\mathbf{x} - \mathbf{y}\|, \end{aligned}$$

by the triangle inequality (Theorem 0.6.4), the Cauchy–Schwarz inequality (Theorem 11.1.5), and Theorem 16.3.2 (a). This is a bound of the kind fixed as the book's continuity test in Chapter 16's introduction. Now let $(w_k)$ be a sequence in $W(\mathbf{A})$, with $w_k = f(\mathbf{x}_k)$ and $\|\mathbf{x}_k\| = 1$. By **the compactness of the unit sphere, fact (A3) of Chapter 16's introduction**, a subsequence $(\mathbf{x}_{k_l})$ converges to a unit vector $\mathbf{x}$. By continuity, $w_{k_l} = f(\mathbf{x}_{k_l}) \to f(\mathbf{x}) \in W(\mathbf{A})$. If $(w_k)$ itself converges to $w$, its subsequence converges to $w$ too, and limits are unique, so $w = f(\mathbf{x}) \in W(\mathbf{A})$: the set is closed.

(e) For a unit $\mathbf{x}$, by the Cauchy–Schwarz inequality and Theorem 16.3.2 (a), $|\mathbf{x}^*\mathbf{A}\mathbf{x}| = |\langle \mathbf{A}\mathbf{x}, \mathbf{x}\rangle| \le \|\mathbf{A}\mathbf{x}\|\,\|\mathbf{x}\| \le \|\mathbf{A}\|_2$.

■

Part (a) makes $W(\mathbf{A})$ an inclusion region for the spectrum, and (e) puts it inside the disc of radius $\|\mathbf{A}\|_2$, which upgrades $\rho(\mathbf{A}) \le \|\mathbf{A}\|_2$ from the spectrum to the whole numerical range. Part (b) is the tool that lets us move any two points of $W(\mathbf{A})$ to 0 and 1, and the next proof depends on it. Part (c) says $W(\mathbf{A})$ depends only on the operator $T_{\mathbf{A}}$ and the inner product, not on the orthonormal basis used to write it down.

## 20.2.3 Convexity: the Toeplitz–Hausdorff theorem

The examples so far were all convex: a point, a segment, a triangle, a disc. That is not a coincidence. To prove it, split $\mathbf{A}$ into two Hermitian pieces, the way a complex number splits into its real and imaginary parts.

**Lemma 20.2.3. Real and Imaginary Parts of a Matrix**

Let $\mathbf{A} \in M_n(\mathbb{C})$, and put

$$\mathbf{H} = \frac{\mathbf{A} + \mathbf{A}^*}{2}, \qquad \mathbf{K} = \frac{\mathbf{A} - \mathbf{A}^*}{2i}.$$

Then $\mathbf{H}$ and $\mathbf{K}$ are Hermitian, $\mathbf{A} = \mathbf{H} + i\mathbf{K}$, and for every $\mathbf{x} \in \mathbb{C}^n$ the numbers $\mathbf{x}^*\mathbf{H}\mathbf{x}$ and

$\mathbf{x}^*\mathbf{K}\mathbf{x}$ are real, with

$$\operatorname{Re}(\mathbf{x}^*\mathbf{A}\mathbf{x}) = \mathbf{x}^*\mathbf{H}\mathbf{x}, \qquad \operatorname{Im}(\mathbf{x}^*\mathbf{A}\mathbf{x}) = \mathbf{x}^*\mathbf{K}\mathbf{x}.$$

*Proof.* By Definition 11.1.2, $(\mathbf{A}^*)^* = \mathbf{A}$, $(\mathbf{B}+\mathbf{C})^* = \mathbf{B}^* + \mathbf{C}^*$, and $(c\mathbf{B})^* = \bar{c}\mathbf{B}^*$. Hence $\mathbf{H}^* = \frac{1}{2}(\mathbf{A}^* + \mathbf{A}) = \mathbf{H}$ and $\mathbf{K}^* = \frac{1}{-2i}(\mathbf{A}^* - \mathbf{A}) = \frac{1}{2i}(\mathbf{A} - \mathbf{A}^*) = \mathbf{K}$. Also $\mathbf{H} + i\mathbf{K} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^*) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^*) = \mathbf{A}$. For $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{x}^*\mathbf{H}\mathbf{x} = \langle \mathbf{H}\mathbf{x}, \mathbf{x}\rangle$ is real by Proposition 11.6.7 (a), applied to the self-adjoint operator $T_{\mathbf{H}}$, and likewise $\mathbf{x}^*\mathbf{K}\mathbf{x}$. So $\mathbf{x}^*\mathbf{A}\mathbf{x} = \mathbf{x}^*\mathbf{H}\mathbf{x} + i\,\mathbf{x}^*\mathbf{K}\mathbf{x}$ is the splitting of a complex number into real and imaginary parts. ∎

For example $\mathbf{J}_2(0)$ has $\mathbf{H} = \frac{1}{2}\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\mathbf{K} = \frac{1}{2i}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = \frac{1}{2}\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, both with eigenvalues $\pm\frac{1}{2}$. A Hermitian $\mathbf{A}$ has $\mathbf{K} = \mathbf{O}$, and then $W(\mathbf{A})$ lies on the real axis.

**Theorem 20.2.4. Toeplitz–Hausdorff Theorem**

For every $\mathbf{A} \in M_n(\mathbb{C})$, the numerical range $W(\mathbf{A})$ is a convex subset of $\mathbb{C}$.

*Idea.* **Step roadmap.** ① By Proposition 20.2.2 (b), an affine change $z \mapsto \alpha z + \beta$ carries any two points of $W(\mathbf{A})$ to 0 and 1, and carries segments to segments; so it suffices to show that $[0,1] \subseteq W(\mathbf{B})$ whenever $0, 1 \in W(\mathbf{B})$. ② Take unit vectors $\mathbf{x}$, $\mathbf{y}$ with $\mathbf{x}^*\mathbf{B}\mathbf{x} = 0$ and $\mathbf{y}^*\mathbf{B}\mathbf{y} = 1$, and walk from $\mathbf{x}$ to $\mathbf{y}$ along the straight path $(1-t)\mathbf{x} + t\mathbf{y}$. Along the way the value of $\mathbf{B}$, normalized, moves continuously from 0 to 1, but it may wander off the real axis. ③ Its imaginary part along the path is $\mathbf{K}$ evaluated there, and expanding shows that only the cross term $\mathbf{x}^*\mathbf{K}\mathbf{y} + \mathbf{y}^*\mathbf{K}\mathbf{x} = 2\operatorname{Re}(\mathbf{x}^*\mathbf{K}\mathbf{y})$ can be nonzero. Multiplying $\mathbf{y}$ by a unimodular scalar rotates $\mathbf{x}^*\mathbf{K}\mathbf{y}$ freely without changing $\mathbf{y}^*\mathbf{B}\mathbf{y}$, so we rotate it onto the imaginary axis, where its real part is 0. ④ Then the path stays on the real axis, and the intermediate value theorem fills in $[0,1]$.

*Proof.* **Step 1: reduction.** Let $w_0, w_1 \in W(\mathbf{A})$ and $0 \le s \le 1$; we must show $(1-s)w_0 + sw_1 \in W(\mathbf{A})$. If $w_0 = w_1$ there is nothing to prove. Otherwise put $\alpha = 1/(w_1 - w_0)$, $\beta = -\alpha w_0$, and $\mathbf{B} = \alpha\mathbf{A} + \beta\mathbf{I}$. By Proposition 20.2.2 (b), $W(\mathbf{B}) = \alpha W(\mathbf{A}) + \beta$, which contains $\alpha w_0 + \beta = 0$ and $\alpha w_1 + \beta = 1$. Suppose we show $s \in W(\mathbf{B})$. Then $s = \alpha w + \beta$ for some $w \in W(\mathbf{A})$, and solving, $w = w_0 + s(w_1 - w_0) = (1-s)w_0 + sw_1$. So it suffices to prove: **if** $0, 1 \in W(\mathbf{B})$**, then** $[0,1] \subseteq W(\mathbf{B})$**.**

**Step 2: choosing the vectors.** Let $\mathbf{B} = \mathbf{H} + i\mathbf{K}$ as in Lemma 20.2.3, and choose unit vectors $\mathbf{x}$, $\mathbf{y}_0$ with $\mathbf{x}^*\mathbf{B}\mathbf{x} = 0$ and $\mathbf{y}_0^*\mathbf{B}\mathbf{y}_0 = 1$. By Lemma 20.2.3, $\mathbf{x}^*\mathbf{H}\mathbf{x} = \mathbf{x}^*\mathbf{K}\mathbf{x} = 0$, $\mathbf{y}_0^*\mathbf{H}\mathbf{y}_0 = 1$, and $\mathbf{y}_0^*\mathbf{K}\mathbf{y}_0 = 0$. Let $c = \mathbf{x}^*\mathbf{K}\mathbf{y}_0$, and put $\gamma = 1$ if $c = 0$, and $\gamma = i\bar{c}/|c|$ otherwise. In both cases $|\gamma| = 1$ and $\gamma c = i|c|$ is purely imaginary. Set $\mathbf{y} = \gamma\mathbf{y}_0$. Then $\|\mathbf{y}\| = 1$ and $\mathbf{y}^*\mathbf{B}\mathbf{y} = |\gamma|^2\,\mathbf{y}_0^*\mathbf{B}\mathbf{y}_0 = 1$, so still $\mathbf{y}^*\mathbf{H}\mathbf{y} = 1$ and $\mathbf{y}^*\mathbf{K}\mathbf{y} = 0$. Since $\mathbf{K}$ is Hermitian, $\mathbf{y}^*\mathbf{K}\mathbf{x} = (\mathbf{x}^*\mathbf{K}\mathbf{y})^* = \overline{\mathbf{x}^*\mathbf{K}\mathbf{y}}$, and so

$$\mathbf{x}^*\mathbf{K}\mathbf{y} + \mathbf{y}^*\mathbf{K}\mathbf{x} = 2\operatorname{Re}(\mathbf{x}^*\mathbf{K}\mathbf{y}) = 2\operatorname{Re}(\gamma c) = 0.$$

Finally, $\mathbf{x}$ and $\mathbf{y}$ are linearly independent. Otherwise, both being nonzero, $\mathbf{y} = \mu\mathbf{x}$ for a scalar $\mu$, and then $1 = \mathbf{y}^*\mathbf{B}\mathbf{y} = |\mu|^2\,\mathbf{x}^*\mathbf{B}\mathbf{x} = 0$, which is absurd.

**Step 3: the path stays real.** For $t \in [0,1]$ put $\mathbf{v}(t) = (1-t)\mathbf{x} + t\mathbf{y}$. The coefficients $1-t$ and $t$ are not both zero, so $\mathbf{v}(t) \neq \mathbf{0}$ by independence. Expanding, and using $\mathbf{x}^*\mathbf{K}\mathbf{x} = \mathbf{y}^*\mathbf{K}\mathbf{y} = 0$ and Step 2,

$$\mathbf{v}(t)^*\mathbf{K}\mathbf{v}(t) = (1-t)^2\,\mathbf{x}^*\mathbf{K}\mathbf{x} + t^2\,\mathbf{y}^*\mathbf{K}\mathbf{y} + t(1-t)\big(\mathbf{x}^*\mathbf{K}\mathbf{y} + \mathbf{y}^*\mathbf{K}\mathbf{x}\big) = 0.$$

So $\mathbf{v}(t)^*\mathbf{B}\mathbf{v}(t) = \mathbf{v}(t)^*\mathbf{H}\mathbf{v}(t)$ is real for every $t$. The same expansion with $\mathbf{H}$ in place of $\mathbf{K}$, and with $\|\mathbf{v}(t)\|^2 = \mathbf{v}(t)^*\mathbf{v}(t)$, gives

$$\begin{aligned} \mathbf{v}(t)^*\mathbf{H}\mathbf{v}(t) &= t^2 + 2t(1-t)\,\mathrm{Re}(\mathbf{x}^*\mathbf{H}\mathbf{y}), \\ \|\mathbf{v}(t)\|^2 &= (1-t)^2 + t^2 + 2t(1-t)\,\mathrm{Re}(\mathbf{x}^*\mathbf{y}), \end{aligned}$$

two polynomials in $t$ with real coefficients, the second positive on $[0,1]$.

**Step 4: filling the segment.** The function

$$f(t) = \frac{\mathbf{v}(t)^*\mathbf{H}\mathbf{v}(t)}{\|\mathbf{v}(t)\|^2}, \qquad 0 \le t \le 1,$$

is a quotient of polynomials with a denominator that never vanishes, hence continuous by the algebra of limits, and $f(0) = \mathbf{x}^*\mathbf{H}\mathbf{x} = 0$, $f(1) = \mathbf{y}^*\mathbf{H}\mathbf{y} = 1$. Let $s \in [0,1]$. By **the intermediate value theorem, fact (A5) of Chapter 16's introduction**, there is $t_s \in [0,1]$ with $f(t_s) = s$. The vector $\mathbf{u} = \mathbf{v}(t_s)/\|\mathbf{v}(t_s)\|$ is a unit vector, and by Step 3,

$$\mathbf{u}^*\mathbf{B}\mathbf{u} = \frac{\mathbf{v}(t_s)^*\mathbf{B}\mathbf{v}(t_s)}{\|\mathbf{v}(t_s)\|^2} = \frac{\mathbf{v}(t_s)^*\mathbf{H}\mathbf{v}(t_s)}{\|\mathbf{v}(t_s)\|^2} = f(t_s) = s.$$

So $s \in W(\mathbf{B})$. By Step 1, $W(\mathbf{A})$ contains the segment between any two of its points, which is Definition 18.1.1. This proves the theorem.

■

The phase choice in Step 2 is the heart of the proof. Without it the imaginary part along the path is $2t(1-t)\,\mathrm{Re}(\mathbf{x}^*\mathbf{K}\mathbf{y}_0)/\|\mathbf{v}(t)\|^2$, which may be nonzero, and the path would leave the real axis between 0 and 1. The rotated vector $\gamma\mathbf{y}_0$ spans the same line as $\mathbf{y}_0$ and gives the same value of $\mathbf{B}$, so the rotation costs nothing. Only the path between $\mathbf{x}$ and $\mathbf{y}$ changes.

With convexity, $W(\mathbf{A})$ contains the convex hull of $\mathrm{spec}(\mathbf{A})$, by (a) and Definition 18.1.4, and it contains the diagonal entries, so it contains $\mathrm{conv}\{a_{11}, \dots, a_{nn}\}$ as well.

## 20.2.4 Normal matrices

For a diagonal matrix the numerical range is the convex hull of the diagonal entries, and those are the eigenvalues. Unitary similarity does not change $W$, and a normal matrix is unitarily similar to a diagonal one. So for a normal matrix $W(\mathbf{A})$ is as small as (a) and convexity allow.

**Proposition 20.2.5. The Numerical Range of a Normal Matrix**

Let $\mathbf{A} \in M_n(\mathbb{C})$ be normal, with eigenvalues $\lambda_1, \dots, \lambda_n$, listed with multiplicity. Then

$$W(\mathbf{A}) = \mathrm{conv}\{\lambda_1, \dots, \lambda_n\}.$$

*Idea.* Unitary similarity leaves $W$ alone, by Proposition 20.2.2 (c), so we may replace $\mathbf{A}$ by a diagonal matrix of its eigenvalues. Then $\mathbf{x}^*\mathbf{A}\mathbf{x}$ is the average of the $\lambda_i$ with weights $|x_i|^2$, and the two inclusions say exactly that the lists of admissible weights and of convex coefficients are the same.

*Proof.* By Corollary 12.4.3, $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$ with $\mathbf{U}$ unitary and $\mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. So $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}$, and $W(\mathbf{A}) = W(\mathbf{D})$ by Proposition 20.2.2 (c). For a unit $\mathbf{x}$, $\mathbf{x}^*\mathbf{D}\mathbf{x} = \sum_i |x_i|^2 \lambda_i$.

($\subseteq$) The weights $|x_i|^2$ are nonnegative with sum 1, so $\mathbf{x}^*\mathbf{D}\mathbf{x}$ is a convex combination of $\lambda_1, \dots, \lambda_n$, and lies in their convex hull by Theorem 18.1.3.

($\supseteq$) By Theorem 18.1.3, a point of the hull is $\sum_{k=1}^{m} t_k \mu_k$ with $t_k \geq 0$, $\sum_k t_k = 1$, and each $\mu_k$ one of the $\lambda_i$. Assign each $k$ to one index $i$ with $\lambda_i = \mu_k$, and let $s_i$ be the sum of the $t_k$ assigned to $i$. Then $s_i \geq 0$, $\sum_i s_i = 1$, and the point is $\sum_i s_i \lambda_i$. The unit vector with $x_i = \sqrt{s_i}$ gives $\mathbf{x}^*\mathbf{D}\mathbf{x} = \sum_i s_i \lambda_i$.

■

For a Hermitian matrix this recovers Proposition 17.1.1. For a unitary matrix, whose eigenvalues lie on the unit circle, $W(\mathbf{A})$ is the convex hull of those eigenvalues: a polygon inscribed in the circle when three or more of them are distinct, and a chord or a single point when there are two or one. For a nonnormal matrix $W(\mathbf{A})$ can be much larger than $\operatorname{conv}\operatorname{spec}(\mathbf{A})$: by Example (The numerical range of a Jordan block), $W(\mathbf{J}_2(0))$ is a disc of radius $\frac{1}{2}$ around the single eigenvalue 0. The extra room is a measure of nonnormality, and Exercise C3 turns it into a comparison with $\|\mathbf{A}\|_2$.

*Quick check.* Find $W(\mathbf{A})$ for the rotation $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, regarded as a complex matrix.

*Solution.* $\mathbf{A}^\top\mathbf{A} = \mathbf{I}$, so $\mathbf{A}$ is real orthogonal, hence unitary and normal. Its characteristic polynomial is $x^2 + 1$, with roots $\pm i$. By Proposition 20.2.5, $W(\mathbf{A})$ is the segment from $-i$ to $i$ on the imaginary axis. For real unit vectors $\mathbf{x}$ the value $\mathbf{x}^*\mathbf{A}\mathbf{x}$ is 0, which is why the real Rayleigh-type quotient of Chapter 17 §01 saw nothing; the complex unit vectors fill in the rest.

## 20.2.5 Bendixson's rectangle

By Lemma 20.2.3, the real part of $\mathbf{x}^*\mathbf{A}\mathbf{x}$ is a value of the Hermitian form of $\mathbf{H}$, and Proposition 17.1.1 confines those values to $[\lambda_n(\mathbf{H}), \lambda_1(\mathbf{H})]$. The same holds for the imaginary part and $\mathbf{K}$. Together they box in the whole numerical range, hence the spectrum.

**Theorem 20.2.6. Bendixson's Theorem**

Let $\mathbf{A} \in M_n(\mathbb{C})$, with $\mathbf{H} = (\mathbf{A} + \mathbf{A}^*)/2$ and $\mathbf{K} = (\mathbf{A} - \mathbf{A}^*)/(2i)$. Every $w \in W(\mathbf{A})$ satisfies

$$\lambda_n(\mathbf{H}) \le \operatorname{Re} w \le \lambda_1(\mathbf{H}),$$
$$\lambda_n(\mathbf{K}) \le \operatorname{Im} w \le \lambda_1(\mathbf{K}).$$

In particular every eigenvalue $\lambda$ of $\mathbf{A}$ satisfies the same inequalities with $\lambda$ in place of $w$.

*Proof.* Let $w = \mathbf{x}^*\mathbf{A}\mathbf{x}$ with $\|\mathbf{x}\| = 1$. By Lemma 20.2.3, $\operatorname{Re} w = \mathbf{x}^*\mathbf{H}\mathbf{x} = \langle \mathbf{H}\mathbf{x}, \mathbf{x}\rangle$ and $\operatorname{Im} w = \mathbf{x}^*\mathbf{K}\mathbf{x}$, and $\mathbf{H}$, $\mathbf{K}$ are Hermitian. By Lemma 13.5.2, $\lambda_n(\mathbf{H}) \le \langle \mathbf{H}\mathbf{x}, \mathbf{x}\rangle \le \lambda_1(\mathbf{H})$, and likewise for $\mathbf{K}$. Every eigenvalue lies in $W(\mathbf{A})$ by Proposition 20.2.2 (a). ∎

The most used consequence concerns the sign of the real parts. If $\mathbf{A} + \mathbf{A}^* \succ \mathbf{O}$, then $\lambda_n(\mathbf{H}) > 0$, so every eigenvalue of $\mathbf{A}$ has positive real part, and in particular $\mathbf{A}$ is invertible. This needs nothing about the size of the off-diagonal entries, only about their symmetric part.

**Example. A rectangle beats the discs**

Let $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ -4 & 1 \end{pmatrix}$. Show that every eigenvalue of $\mathbf{A}$ has positive real part, using Theorem 20.2.6, and check that the Gershgorin discs cannot show it. Compare with the eigenvalues.

*Solution.* Here $\mathbf{A}^* = \mathbf{A}^{\top}$, so

$$\mathbf{H} = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \mathbf{K} = \frac{1}{2i}\begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -3i \\ 3i & 0 \end{pmatrix}.$$

$\mathbf{H}$ has trace $4$ and determinant $2$, so $p_{\mathbf{H}}(x) = x^2 - 4x + 2$, with roots $2 \pm \sqrt{2}$. $\mathbf{K}$ has trace $0$ and determinant $0 - (-3i)(3i) = -9$, so its eigenvalues are $\pm 3$. By Theorem 20.2.6, every eigenvalue lies in the rectangle

$$2 - \sqrt{2} \le \operatorname{Re}\lambda \le 2 + \sqrt{2}, \qquad -3 \le \operatorname{Im}\lambda \le 3,$$

and in particular $\operatorname{Re}\lambda \ge 2 - \sqrt{2} > 0$.

The row discs are $|z - 3| \le 2$ and $|z - 1| \le 4$, and the second contains $-3$. The column discs are $|z - 3| \le 4$ and $|z - 1| \le 2$, and the first contains $-1$. So neither union rules out an eigenvalue with negative real part.

Directly, $p_{\mathbf{A}}(x) = x^2 - 4x + 11$, so $\lambda = 2 \pm i\sqrt{7}$. The real part $2$ lies in $[2 - \sqrt{2}, 2 + \sqrt{2}]$, and the imaginary parts $\pm\sqrt{7} \approx \pm 2.65$ lie in $[-3, 3]$.

## 20.2.6 What these regions do not do

Every region in this section and the last is an **inclusion region**: it contains the spectrum. None of them is a perturbation bound in the sense of the rest of this chapter, which asks how far the eigenvalues move when $\mathbf{A}$ changes by a small $\mathbf{E}$. The numerical range does give a statement of

that kind. For a unit $\mathbf{x}$, $\mathbf{x}^*(\mathbf{A}+\mathbf{E})\mathbf{x} = \mathbf{x}^*\mathbf{A}\mathbf{x}+\mathbf{x}^*\mathbf{E}\mathbf{x}$, and $|\mathbf{x}^*\mathbf{E}\mathbf{x}| \le \|\mathbf{E}\|_2$ by Proposition 20.2.2 (e). Hence

$$W(\mathbf{A}+\mathbf{E}) \subseteq W(\mathbf{A}) + W(\mathbf{E}) \subseteq \{\, w+e : w \in W(\mathbf{A}),\ |e| \le \|\mathbf{E}\|_2 \,\},$$

and so every eigenvalue of $\mathbf{A}+\mathbf{E}$ lies within $\|\mathbf{E}\|_2$ of $W(\mathbf{A})$. This is only as good as $W(\mathbf{A})$ is small.

*Warning.* **$W(\mathbf{A})$ can be far larger than $\operatorname{spec}(\mathbf{A})$, so "within $\|\mathbf{E}\|_2$ of $W(\mathbf{A})$" is not "within $\|\mathbf{E}\|_2$ of an eigenvalue".** For $\mathbf{A} = M\mathbf{J}_2(0)$ with $M > 0$, Proposition 20.2.2 (b) and Example (The numerical range of a Jordan block) give $W(\mathbf{A}) = \{|z| \le M/2\}$, while $\operatorname{spec}(\mathbf{A}) = \{0\}$. So the bound above allows the eigenvalues of $\mathbf{A}+\mathbf{E}$ to lie anywhere in $|z| \le M/2 + \|\mathbf{E}\|_2$, and this does not shrink as $\mathbf{E} \to \mathbf{O}$. The eigenvalues do move a long way: $\mathbf{A} + \varepsilon \mathbf{e}_2\mathbf{e}_1^{\top}$ has characteristic polynomial $x^2 - M\varepsilon$, so for $\varepsilon > 0$ its eigenvalues $\pm\sqrt{M\varepsilon}$ have moved by $\sqrt{M\varepsilon}$, far more than $\varepsilon$. For a normal $\mathbf{A}$, $W(\mathbf{A}) = \operatorname{conv}\operatorname{spec}(\mathbf{A})$ is small, but it still contains points far from every eigenvalue, such as the midpoint of two distant ones.

The genuine perturbation bounds need a different mechanism: §03 proves that eigenvalues move continuously and at what rate in general, and §04 gives Bauer–Fike's bound, in which the distance to the nearest **eigenvalue** is controlled.

## 20.2.7 Exercises

### 20.2.7.1 A. Check your understanding

**Exercise. A1**

(a) Define the Brauer oval $O_{ij}(\mathbf{A})$ and the numerical range $W(\mathbf{A})$, and state the Toeplitz–Hausdorff theorem.

(b) True or false: $W(\mathbf{A}) = \operatorname{conv}\operatorname{spec}(\mathbf{A})$ for every $\mathbf{A} \in M_n(\mathbb{C})$. Justify your answer.

(c) True or false: $W(\mathbf{A})$ is contained in the union of the Gershgorin discs of $\mathbf{A}$. Justify your answer.

(d) True or false: if $\mathbf{A} + \mathbf{A}^* \succ \mathbf{O}$, then $\mathbf{A}$ is invertible. Justify your answer.

(e) Why does Theorem 20.2.1 require $n \ge 2$?

*Solution.*

(a) For $i \ne j$, $O_{ij}(\mathbf{A}) = \{z : |z - a_{ii}||z - a_{jj}| \le r_i(\mathbf{A})r_j(\mathbf{A})\}$, and $W(\mathbf{A}) = \{\mathbf{x}^*\mathbf{A}\mathbf{x} : \|\mathbf{x}\| = 1\}$. Theorem 20.2.4: for every square complex $\mathbf{A}$, $W(\mathbf{A})$ is convex.

(b) False. $W(\mathbf{J}_2(0))$ is the disc $|z| \le \frac{1}{2}$ (Example (The numerical range of a Jordan block)), while $\operatorname{conv}\operatorname{spec}(\mathbf{J}_2(0)) = \{0\}$. Equality holds for normal matrices (Proposition 20.2.5).

(c) False. For $\mathbf{A} = \operatorname{diag}(1, -1)$ the discs are the two points $\{1\}$ and $\{-1\}$, while $W(\mathbf{A}) = [-1, 1]$ contains 0. The discs contain the spectrum, not the numerical range.

(d) True. By Theorem 20.2.6 every eigenvalue has $\operatorname{Re}\lambda \ge \lambda_n(\mathbf{H}) > 0$, with $\mathbf{H} = (\mathbf{A} + \mathbf{A}^*)/2 \succ \mathbf{O}$. So 0 is not an eigenvalue, and $\mathbf{A}$ is invertible (Theorem 9.2.1).

(e) An oval is built from two distinct indices $i \ne j$, and the proof needs a second index $q \ne p$. For $n = 1$ there are no ovals, and the union in (a) would be empty, although

the matrix has an eigenvalue.

#### 20.2.7.2 B. Practice

**Exercise. B1: Invertibility from ovals**

Let $\mathbf{A} = \begin{pmatrix} 5 & 1 & 1 \\ 1 & 5 & 1 \\ 2 & 2 & 2 \end{pmatrix}$.

(a) Show that neither the row discs nor the column discs of $\mathbf{A}$ exclude 0.

(b) Use Theorem 20.2.1 to show that $\mathbf{A}$ is invertible.

(c) Check that $(1, -1, 0)$ is an eigenvector, and verify that its eigenvalue lies in $O_{12}(\mathbf{A})$.

*Solution.*

(a) The row radii are $2, 2, 4$, and $D_3(\mathbf{A}) = \{|z - 2| \le 4\}$ contains 0. The column radii are $3, 3, 2$, and the third column disc $|z - 2| \le 2$ contains 0.

(b) At $z = 0$: for $O_{12}$, $5 \cdot 5 = 25 > 4 = r_1 r_2$; for $O_{13}$ and $O_{23}$, $5 \cdot 2 = 10 > 8 = r_1 r_3 = r_2 r_3$. So 0 lies in no oval, is not an eigenvalue by Theorem 20.2.1 (a), and $\mathbf{A}$ is invertible. (Its determinant is 32.)

(c) $\mathbf{A}(1, -1, 0) = (5 - 1, 1 - 5, 2 - 2) = (4, -4, 0)$, so the eigenvalue is 4. For $O_{12}$: $|4 - 5||4 - 5| = 1 \le 4$.

**Exercise. B2: A Bendixson rectangle**

Let $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ -1 & 1 \end{pmatrix}$. Find the rectangle of Theorem 20.2.6, and hence show that every eigenvalue of $\mathbf{A}$ satisfies $0 \le \operatorname{Re}\lambda \le 2$ and $|\operatorname{Im}\lambda| \le 2$. Compare with the eigenvalues.

*Solution.* $\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with $p_{\mathbf{H}}(x) = x^2 - 2x$, so its eigenvalues are 0 and 2. $\mathbf{K} = \frac{1}{2i}(\mathbf{A} - \mathbf{A}^\top) = \frac{1}{2i}\left(\begin{smallmatrix} 0 & 4 \\ -4 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & -2i \\ 2i & 0 \end{smallmatrix}\right)$, with trace 0 and determinant $-4$, so its eigenvalues are $\pm 2$. By Theorem 20.2.6 every eigenvalue has $0 \le \operatorname{Re}\lambda \le 2$ and $-2 \le \operatorname{Im}\lambda \le 2$. Directly, $p_{\mathbf{A}}(x) = x^2 - 2x + 4$, so $\lambda = 1 \pm i\sqrt{3}$: the real part 1 lies in $[0, 2]$, and $\sqrt{3} \approx 1.73 \le 2$.

**Exercise. B3: Computing numerical ranges**

Determine $W(\mathbf{A})$ for each matrix. Justify your answers.

(a) $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

(b) $\mathbf{A} = \operatorname{diag}(1, i, -1, -i)$

(c) $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

*Solution.*

(a) $\mathbf{A} = 2\mathbf{J}_2(0) + \mathbf{I}$, so by Proposition 20.2.2 (b) and Example (The numerical range of a Jordan block), $W(\mathbf{A}) = 2W(\mathbf{J}_2(0)) + 1 = \{z : |z - 1| \le 1\}$, the closed disc of radius 1 about 1.

(b) A diagonal matrix is normal, so by Proposition 20.2.5 $W(\mathbf{A}) = \operatorname{conv}\{1, i, -1, -i\}$, the filled square with these four vertices.

(c) $\mathbf{A}$ is real symmetric, with eigenvalues 3 and 1 (eigenvectors $(1, 1)$ and $(1, -1)$). By Proposition 17.1.1, or Proposition 20.2.5, $W(\mathbf{A}) = [1, 3]$.

### 20.2.7.3 C. Going deeper

**Exercise. C1: Direct sums**

(a) Let $\mathbf{B} \in M_k(\mathbb{C})$ and $\mathbf{C} \in M_m(\mathbb{C})$. Prove that $W(\mathbf{B} \oplus \mathbf{C}) = \operatorname{conv}(W(\mathbf{B}) \cup W(\mathbf{C}))$.

(b) Deduce that $\mathbf{A} = \operatorname{diag}(2, -2, 2i, -2i) \oplus \mathbf{J}_2(0)$ is not normal, but satisfies $W(\mathbf{A}) = \operatorname{conv} \operatorname{spec}(\mathbf{A})$.

*Hint for (a): one inclusion needs Theorem 20.2.4.*

*Solution.*

(a) ($\subseteq$) Write a unit vector of $\mathbb{C}^{k+m}$ as $\mathbf{x} = (\mathbf{y}, \mathbf{z})$, with $\|\mathbf{y}\|^2 + \|\mathbf{z}\|^2 = 1$. Then $\mathbf{x}^*(\mathbf{B} \oplus \mathbf{C})\mathbf{x} = \mathbf{y}^*\mathbf{B}\mathbf{y} + \mathbf{z}^*\mathbf{C}\mathbf{z}$. If $\mathbf{z} = \mathbf{0}$ this is a point of $W(\mathbf{B})$, and if $\mathbf{y} = \mathbf{0}$ a point of $W(\mathbf{C})$. Otherwise, with $s = \|\mathbf{y}\|^2 \in (0, 1)$,

$$\mathbf{x}^*(\mathbf{B} \oplus \mathbf{C})\mathbf{x} = s\,\frac{\mathbf{y}^*\mathbf{B}\mathbf{y}}{\|\mathbf{y}\|^2} + (1 - s)\,\frac{\mathbf{z}^*\mathbf{C}\mathbf{z}}{\|\mathbf{z}\|^2},$$

a convex combination of a point of $W(\mathbf{B})$, at the unit vector $\mathbf{y}/\|\mathbf{y}\|$, and a point of $W(\mathbf{C})$. In every case the value lies in $\operatorname{conv}(W(\mathbf{B}) \cup W(\mathbf{C}))$, by Theorem 18.1.3.
($\supseteq$) Taking $\mathbf{x} = (\mathbf{y}, \mathbf{0})$ with $\mathbf{y}$ a unit vector shows $W(\mathbf{B}) \subseteq W(\mathbf{B} \oplus \mathbf{C})$, and likewise $W(\mathbf{C}) \subseteq W(\mathbf{B} \oplus \mathbf{C})$. By Theorem 20.2.4, $W(\mathbf{B} \oplus \mathbf{C})$ is convex, so it contains the smallest convex set containing $W(\mathbf{B}) \cup W(\mathbf{C})$ (Definition 18.1.4).

(b) $\mathbf{A}$ is not normal: its lower-right block is $\mathbf{J}_2(0)$, and $\mathbf{A}^*\mathbf{A}$ and $\mathbf{A}\mathbf{A}^*$ differ in that block, since $\mathbf{J}_2(0)^*\mathbf{J}_2(0) = \operatorname{diag}(0, 1)$ and $\mathbf{J}_2(0)\mathbf{J}_2(0)^* = \operatorname{diag}(1, 0)$. By (a), Proposition 20.2.5, and Example (The numerical range of a Jordan block), $W(\mathbf{A})$ is the convex hull of the square $Q = \operatorname{conv}\{\pm 2, \pm 2i\}$ and the disc $\{|z| \le \frac{1}{2}\}$. The square contains the disc: $Q = \{z : |\operatorname{Re} z| + |\operatorname{Im} z| \le 2\}$, and $|\operatorname{Re} z| + |\operatorname{Im} z| \le 2|z| \le 1$ on the disc. So $W(\mathbf{A}) = Q$. The spectrum is $\{\pm 2, \pm 2i, 0\}$, whose hull is also $Q$, since $0 \in Q$. Hence $W(\mathbf{A}) = \operatorname{conv} \operatorname{spec}(\mathbf{A})$, although $\mathbf{A}$ is not normal: the converse of Proposition 20.2.5 fails.

**Exercise. C2: When the numerical range is real, or a point**

Let $\mathbf{A} \in M_n(\mathbb{C})$.

(a) Prove that $W(\mathbf{A}) \subseteq \mathbb{R}$ if and only if $\mathbf{A}$ is Hermitian.

(b) Deduce that $W(\mathbf{A}) = \{c\}$ for a single $c \in \mathbb{C}$ if and only if $\mathbf{A} = c\mathbf{I}$.

*Solution.*

(a) ($\Leftarrow$) If $\mathbf{A}$ is Hermitian, $\mathbf{x}^*\mathbf{A}\mathbf{x} = \langle \mathbf{A}\mathbf{x}, \mathbf{x}\rangle$ is real by Proposition 11.6.7 (a). ($\Rightarrow$) Write $\mathbf{A} = \mathbf{H} + i\mathbf{K}$ as in Lemma 20.2.3. If $W(\mathbf{A}) \subseteq \mathbb{R}$, then $\mathbf{x}^*\mathbf{K}\mathbf{x} = \operatorname{Im}(\mathbf{x}^*\mathbf{A}\mathbf{x}) = 0$ for every unit $\mathbf{x}$. By Proposition 17.1.1, $\lambda_1(\mathbf{K})$ and $\lambda_n(\mathbf{K})$ are values of $\mathbf{x}^*\mathbf{K}\mathbf{x}$ on unit vectors, so both are 0, and every eigenvalue of $\mathbf{K}$ is 0. By Corollary 12.4.3, $\mathbf{K} = \mathbf{U}\,\mathbf{O}\,\mathbf{U}^* = \mathbf{O}$. Hence $\mathbf{A} = \mathbf{H}$ is Hermitian.

(b) ($\Leftarrow$) is the example $W(c\mathbf{I}) = \{c\}$. ($\Rightarrow$) By Proposition 20.2.2 (b), $W(\mathbf{A} - c\mathbf{I}) = W(\mathbf{A}) - c = \{0\} \subseteq \mathbb{R}$, so $\mathbf{A} - c\mathbf{I}$ is Hermitian by (a). By Proposition 17.1.1 its extreme eigenvalues are values of the form on unit vectors, hence both 0, and as in (a) the Hermitian matrix $\mathbf{A} - c\mathbf{I}$ with all eigenvalues 0 is $\mathbf{O}$.

**Exercise. C3: The numerical radius**

For $\mathbf{A} \in M_n(\mathbb{C})$, the **numerical radius** is $w(\mathbf{A}) = \max\{|z| : z \in W(\mathbf{A})\}$.

(a) Explain why the maximum exists, and prove that $\rho(\mathbf{A}) \le w(\mathbf{A}) \le \|\mathbf{A}\|_2$.

(b) Prove that $\|\mathbf{A}\|_2 \le 2\,w(\mathbf{A})$.

(c) Show that both constants are attained: $w(\mathbf{A}) = \|\mathbf{A}\|_2$ for every normal $\mathbf{A}$, and $\|\mathbf{A}\|_2 = 2\,w(\mathbf{A})$ for $\mathbf{A} = \mathbf{J}_2(0)$.

*Hint for (b): bound $\|\mathbf{H}\|_2$ and $\|\mathbf{K}\|_2$ separately.*

*Solution.*

(a) $W(\mathbf{A})$ is nonempty and compact (Proposition 20.2.2 (d)), and $z \mapsto |z|$ is continuous, so the maximum is attained by **the extreme value theorem, fact (A4) of Chapter 16's introduction**. Every eigenvalue lies in $W(\mathbf{A})$, by Proposition 20.2.2 (a), so $\rho(\mathbf{A}) \le w(\mathbf{A})$; and $w(\mathbf{A}) \le \|\mathbf{A}\|_2$ by Proposition 20.2.2 (e).

(b) Let $\mathbf{A} = \mathbf{H} + i\mathbf{K}$ as in Lemma 20.2.3. By Lemma 17.3.5, $\|\mathbf{H}\|_2$ is the larger of $\lambda_1(\mathbf{H})$ and $-\lambda_n(\mathbf{H})$. By Proposition 17.1.1 there is a unit $\mathbf{x}$ with

$$\lambda_1(\mathbf{H}) = \mathbf{x}^*\mathbf{H}\mathbf{x} = \operatorname{Re}(\mathbf{x}^*\mathbf{A}\mathbf{x}) \le |\mathbf{x}^*\mathbf{A}\mathbf{x}| \le w(\mathbf{A}),$$

and a unit $\mathbf{y}$ with $-\lambda_n(\mathbf{H}) = -\operatorname{Re}(\mathbf{y}^*\mathbf{A}\mathbf{y}) \le w(\mathbf{A})$. So $\|\mathbf{H}\|_2 \le w(\mathbf{A})$, and the same argument with imaginary parts gives $\|\mathbf{K}\|_2 \le w(\mathbf{A})$. As the operator norm is a norm (Theorem 16.3.2 (b)), $\|\mathbf{A}\|_2 \le \|\mathbf{H}\|_2 + \|i\mathbf{K}\|_2 \le 2\,w(\mathbf{A})$.

(c) For normal $\mathbf{A}$, $\rho(\mathbf{A}) = \|\mathbf{A}\|_2$, as Chapter 16 §04 showed after Definition 16.4.1, so the chain in (a) collapses to equalities. For $\mathbf{J}_2(0)$, Example (The numerical range of a Jordan block) gives $w(\mathbf{J}_2(0)) = \frac{1}{2}$, and $\|\mathbf{J}_2(0)\|_2 = 1$, since $\|\mathbf{J}_2(0)\mathbf{x}\| = |x_2| \le \|\mathbf{x}\|$ with equality at $\mathbf{x} = \mathbf{e}_2$. So $\|\mathbf{J}_2(0)\|_2 = 2\,w(\mathbf{J}_2(0))$.